

FinMitra — Project Context & PPT Blueprint

An Agentic AI Financial Adviser & Personal Wealth Management Platform

EXECUTIVE SUMMARY: FinMitra bridges the gap between passive expense trackers and personal wealth advising. By pairing a high-performance Java Spring Boot REST API with a context-aware Agentic AI engine powered by Google Gemini, FinMitra analyzes real-time user cash flow, budget limits, and savings telemetry to deliver actionable, personalized financial advice.

1. The Problem Statement & Industry Gaps

Personal finance apps have existed for over a decade, yet young professionals and retail users continue to struggle with debt, impulse spending, and poor wealth accumulation. FinMitra directly addresses four major industry problems:

- **Passive Data vs. Actionable Advice:** Traditional apps display static graphs showing where money went ('You spent ₹15,000 on dining out'), but fail to advise users on what to do next.
- **Financial Blindspots:** Many individuals track expenses in scattered notes or spreadsheets without understanding their net monthly cash flow or safe spending thresholds.
- **Uninformed Loan & EMI Decisions:** Users frequently buy gadgets, cars, or take loans without knowing if their net monthly savings can comfortably absorb the EMI without causing financial distress.
- **Generic LLM Limitations:** Standard public AI chatbots lack access to the user's real bank balance and category budgets, leading to generic, ineffective financial responses.

2. What We Are Developing (Core Features)

Module	Feature Description	Key Capabilities
User Auth & Security	Stateless JWT authentication system with BCrypt password encryption.	User registration, secure login, bearer token interceptors.
Transaction Ledger	Full CRUD expense/income ledger linked to MySQL backend.	Categorization (Rent, Food, Transport, etc.), date sorting, transaction notes.
Budget & Alerts System	Category spending caps with real-time threshold monitoring.	Visual progress bars, over-budget warning badges, remaining cap calculation.
Visual Analytics UI	Interactive financial dashboard built with React 18 & Recharts.	Category Donut charts, Income vs Expense bar comparison, Net Savings KPIs.
Mitra AI Adviser	Context-aware Agentic AI financial engine powered by Google Gemini.	Monthly summary generation, safe EMI calculator, SIP strategy, Q&A; chatbot.
Dual Theme System	1-click toggle between Dark Obsidian and Warm Cream Day UI.	Custom CSS design tokens, smooth color palette transition.

3. Tech Stack & Architectural Justification

Technology	Component Role	Why We Chose It (Rationale)
Java Spring Boot 3.3.2	Backend REST API	Enterprise-grade reliability, high performance, structured layered architecture (Controller, Service, Repository, DTO).
MySQL 9.3.0 / Supabase	Relational Database	ACID compliance, strict foreign key constraints (ON DELETE CASCADE), indexed queries for fast financial calculations.
Spring Security + JWT	Security & Auth	Stateless Bearer Token security model. Passwords hashed using BCrypt to satisfy financial data safety standards.
Google Gemini 2.5 Flash	Agentic AI Engine	Ultra-low latency, high reasoning accuracy, cost-effective API inference, native support for injected RAG database telemetry.
React 18 + Vite 5	Frontend Web App	Lightning-fast HMR dev environment with Vite, component-driven UI architecture with React Hooks & Context API.
Recharts & Lucide Icons	Data Viz & Graphics	Lightweight, responsive SVG graphing primitives tailored for financial web applications.

4. Agentic AI & Database Telemetry Engine

- The core innovation of FinMitra lies in its **Live Telemetry Injector** implemented in AIServiceImpl.java. Unlike standard chatbots, FinMitra dynamically constructs an in-memory financial profile before querying the AI model:
- **Real-time Aggregation:** Calculates total income, total expense, net savings, and category distribution directly from MySQL database tables.
 - **Context Injection:** Generates a structured system prompt containing exact financial metrics: *'User: Admin, Income: █50,000, Expense: █28,000, Net Savings: █22,000, Highest Spend Category: Rent (█12,000)'*.
 - **Safe Loan & EMI Evaluation:** When asked about loans or purchases, the engine evaluates if monthly EMI stays under 35% of net savings: $\text{Max Safe EMI} = \text{Net Savings} \times 0.35$.
 - **Wealth Growth Allocation:** Automatically calculates 30% of net savings for SIPs or liquid mutual funds.
 - **Deterministic Fallback Engine:** If external AI API calls fail or offline, a rules-based fallback engine guarantees 100% operational response uptime.

5. 10-Slide PPT Presentation Blueprint

Use the following structured outline to quickly populate your project presentation slides:

Slide 1: Title & Introduction

Title: FinMitra — An Agentic AI Finance Adviser
Subtitle: Smart Personal Finance & Wealth Management Platform
Visual: High-tech dark obsidian mockup with green accent glows.

Slide 2: Executive Summary

Headline: Bridging the Gap Between Ledger Logging and Wealth Advice
Key Message: Full-stack platform combining Java Spring Boot, MySQL, React, and Google Gemini AI telemetry.

Slide 3: The Problem Statement

Headline: Why Existing Budget Apps Fail Young Professionals
Key Points: Data without action, unmonitored impulse spending, risky loan/EMI choices, generic AI chatbots.

Slide 4: The FinMitra Solution

Headline: Actionable Financial Intelligence Powered by AI Telemetry
Key Points: Real-time net cash flow tracking, automated category budget alerts, context-aware AI financial advice.

Slide 5: System Architecture

Headline: Enterprise Layered Architecture
Key Components: React 18 Frontend -> Spring Boot REST API -> Spring Security JWT -> MySQL Database -> Google Gemini API.

Slide 6: Agentic AI Telemetry Engine

Headline: How Mitra AI Delivers Personal Advice
Key Features: In-memory cash flow synthesis, 35% net savings EMI threshold safety rule, 30% SIP wealth allocation engine.

Slide 7: Tech Stack Matrix

Headline: Production-Grade Stack Choice & Rationale
Matrix: Spring Boot (Security & Scale), MySQL (ACID Data Safety), Gemini 2.5 (Fast AI Inference), React+Vite (Modern UX).

Slide 8: Key UI & User Experience

Headline: Designed for Clarity and Engagement
Features: Sleek Dark Obsidian / Warm Cream themes, Recharts visual analytics, interactive chat widget.

Slide 9: Project Status & Achievements

Headline: Completed Modules & Validation
Milestones: ■ Auth & JWT, ■ Transaction CRUD, ■ Category Budgets, ■ Analytics Dashboard, ■ Gemini AI Engine.

Slide 10: Future Roadmap & Vision

Headline: Scaling FinMitra to Next-Gen Wealth Platform
Vision: Bank SMS Auto-parsing, PDF/CSV Tax Export, Goal Vaults, Multi-currency & Regional Indian Language support.

6. Future Scope & Strategic Roadmap

- **Account Aggregator (AA) Integration:** Automate transaction logging via bank SMS feeds and official Account Aggregator APIs in India.
- **PDF & CSV Export Engine:** Allow users to generate audited monthly financial reports, expense breakdowns, and tax summaries using OpenPDF / Apache POI.
- **Goal-Based Savings Vaults:** Enable users to create specific financial targets (e.g. Emergency Fund, Vehicle Purchase) with automatic savings allocation.
- **Multi-Currency & Regional Localization:** Expand beyond INR (₹) to USD (\$) / EUR (€) and add Indian regional languages (Hindi, Marathi, Tamil, Telugu).
- **Mobile Native App:** Extend web application to mobile platforms using React Native with push notifications for budget limit breaches.